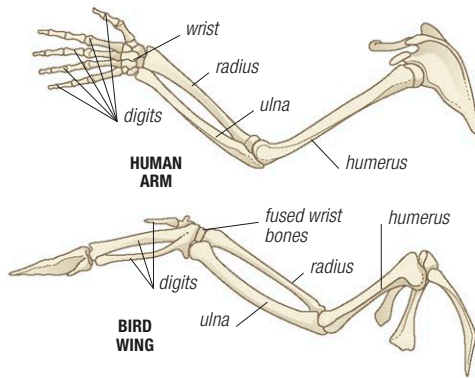




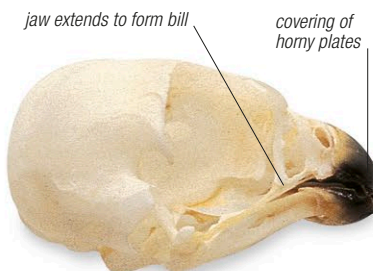
ARM AND WING BONES

In comparison with a human arm, a bird's forelimb (or wing) has undergone a dramatic reduction in the number of bones, especially around the wrist and hand.

weight, a number of bones are greatly reduced in size, and many have fused together, making a rigid frame without the need for large muscles and ligaments to hold the bones together. Most birds' bones are also hollow, lacking marrow. To compensate for their lightness, they are supported by a network of internal struts (trabeculae) at points of stress, giving them great strength. Several bones, including the humerus, contain air sacs, which are connected to the respiratory system.

A bird's wing is the anatomical counterpart of the human arm, although birds have relatively few digits, and some of the "hand" bones are fused together, which contributes to the general rigidity of the structure. Similarly, the "elbow" and "wrist" joints of the wing bones are inflexible in the vertical plane. As a result, when the bird is flying, the wings are held straight out and flapped only at the shoulder, providing rigidity and saving energy by preventing unnecessary movement. The powerful wing muscles are attached to a keel, a large projection that extends at right angles from the breastbone.

Another feature unique to birds is their light, flexible bill (or beak). In some birds, such as parrots, both jaws (or mandibles) move, giving a wider gape. In mammals, only the lower jaw moves. The bill itself has an external covering of keratin, which is light and strong and allows for great variety in form. The bill shape is always adapted to a bird's particular method of feeding, and reflects its diet—for example, finches have a strong,

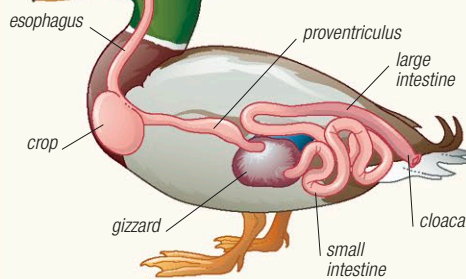


SKULL

The skull of a bird is very light, with many of the bones fused together. The bill consists of an upper and lower mandible, with a highly variable horny sheath on both.

DIGESTIVE SYSTEM

Since they lack teeth, birds are forced to break up particles of food farther down the alimentary canal, in the stomach. The lower part of the stomach (gizzard) grinds down material, often aided by an abrasive paste containing ingested grit or stones, while the upper part (proventriculus) secretes gastric juices. The food is often stored in the esophagus for later assimilation. In some birds, the esophagus expands near its base to make a saclike crop, a further storage vessel. Food can be transferred there very quickly, allowing for large quantities to be ingested in a relatively short time. This is valuable for birds that must risk exposure to predators while feeding.



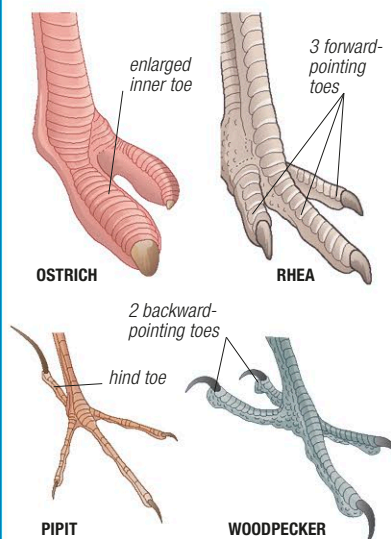
FEEDING AND STORING FOOD

A bird's digestive system is adapted for a dynamic lifestyle. Food is ingested without chewing, and can be stored in the esophagus or the crop. This allows the bird to feed quickly, but to digest in a safer place.



CROP FEEDING

Adult pigeons are unusual in producing a milky secretion from the crop to feed to their young. Here, a turtle dove gives crop milk to a nestling.



FEET AND TOES

Ostriches and rheas have fewer toes than most birds, enabling them to run quickly. The pipit has the perching foot typical of passerines, with 3 toes pointing forward and 1 back. The arrangement of a woodpecker's toes allows it to climb easily in all directions.

conical bill for cracking seeds, whereas herons have a pointed, daggerlike bill for seizing fish. Birds that feed on insects, such as warblers, have a small, slender bill, while in flesh-eaters, such as hawks, the bill is sharp and hooked for tearing open prey.

Birds have between 2 and 4 toes on their feet, the arrangement and shape varying according to the bird's way of life. Most have 4 toes, of which 3 point forward and one back, a good system for perching (see p.342). Many swimming birds have webs between their toes.

Like humans, birds rely primarily on their sense of sight, followed by hearing. Indeed, many birds have

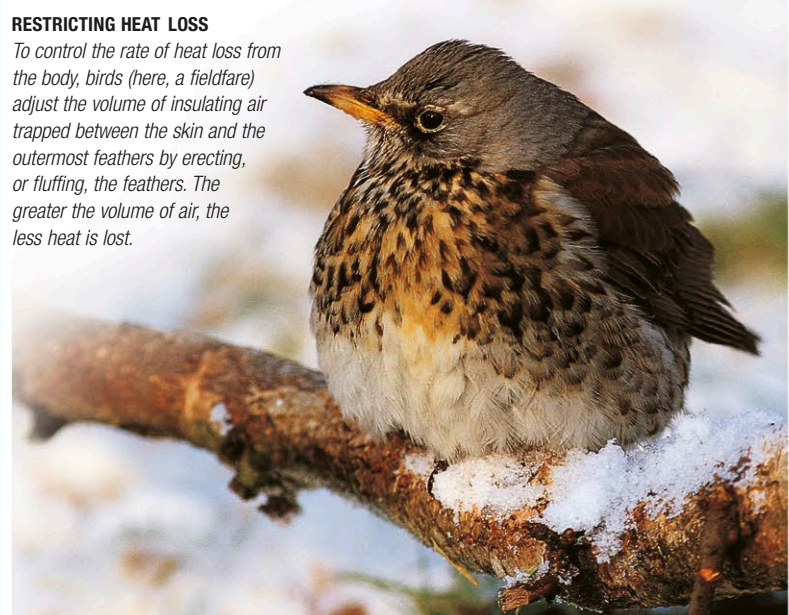
exceptional vision, particularly owls and birds of prey. With their high-speed aerial lifestyle, birds rarely (and in some cases, never) use their sense of smell.

Respiration and circulation

Birds are active animals that have a high metabolic rate. They have an efficient respiratory system that extracts large amounts of oxygen from the air, and a circulation system that can move the oxygen rapidly around the body. Efficient oxygen extraction is also necessary so that birds can remain active at high altitude, where oxygen is less abundant. Although the lungs are small, they are connected to a series of air sacs found throughout the

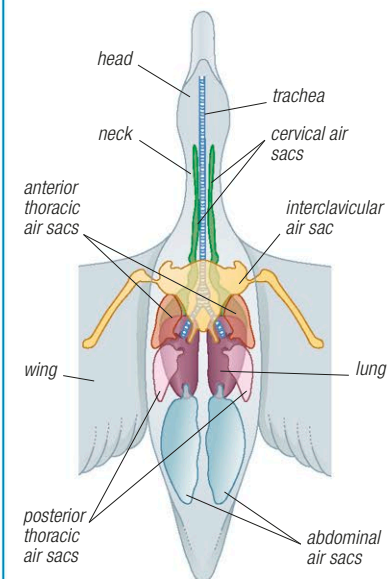
RESTRICTING HEAT LOSS

To control the rate of heat loss from the body, birds (here, a fieldfare) adjust the volume of insulating air trapped between the skin and the outermost feathers by erecting, or fluffing, the feathers. The greater the volume of air, the less heat is lost.



body, which help inflate and deflate the lungs. Instead of air flowing alternately in and out, as it does in mammals, it flows in a single direction. Working in tandem with this efficient oxygen-extraction system, birds have a large heart that pumps at a relatively rapid rate.

Birds need to maintain a constant internal temperature of about 104°F (40°C) to remain active. To achieve this, they control their rate of heat loss (see below). In some species, when there is not enough energy to sustain the metabolism, the internal temperature falls and the bird enters an inactive state called torpor. Some birds that live at high altitude, such as certain hummingbirds, become torpid overnight; others, such as some swifts and nightjars, can stay in this state for days or even weeks.



AIR SACS

A bird's lungs are only part of its air-intake system. Connected to the lungs is a system of air sacs, which have thin, nonmuscular walls and perform a similar function to the diaphragm in humans. In all, there are 9 air sacs throughout the body, 8 of which are arranged in pairs.